Electrician Contractor Management System

Project Report

1. Project Description

The **Electrician Contractor Management System (ECMS)** is a full-stack web application developed to efficiently manage electricians, contractors, jobs, and tasks within an organization.

The system provides a centralized platform where administrators and contractors can assign jobs and tasks, while electricians can track their work, update task status, and receive notifications.

The application enhances productivity by automating task assignment, tracking deadlines, and providing real-time updates through a notification system.

2. Features and Technologies

♦ Core Features

- **Authentication System**
  - JWT-based login and registration
  - Role-based access (Admin, Contractor, Electrician)
- **Electrician Management**
  - Add, update, and manage electricians
  - Assign tasks and jobs
- **Job Management**
  - Create and assign jobs
  - Track job details and deadlines
- **Task Management**
  - Add, edit, delete tasks
  - Assign tasks to electricians
  - Track task status (Pending, In Progress, Completed)
- **Notification System**
  - Task assignment notifications
  - Task update notifications
  - Task completion notifications (Admin)
  - Deadline alerts
- **Dashboard**
  - Role-based dashboard view
  - Task statistics (Pending, Completed, etc.)
  - Completion rate visualization

- **Reports**
    - Overview of system performance
    - Task and job analytics

♦ Technologies Used

Backend

- Django
- Django REST Framework
- SimpleJWT (Authentication)

Frontend

- HTML
- CSS
- Bootstrap
- JavaScript

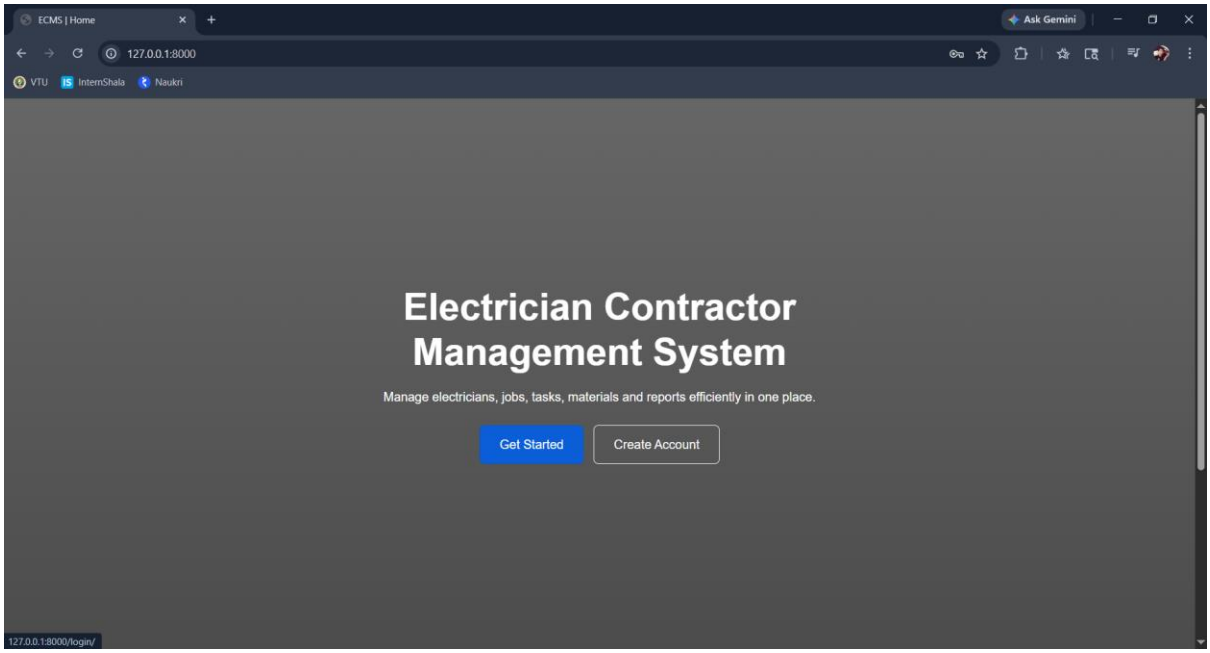
Database

- SQLite

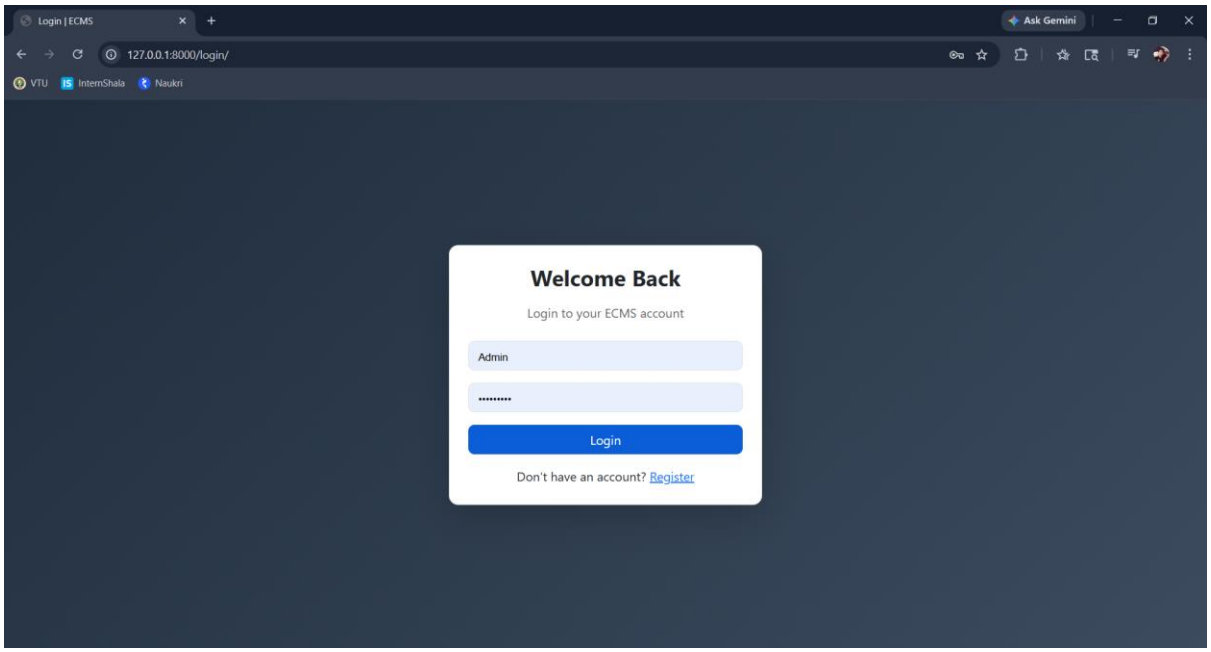
3. Screenshots

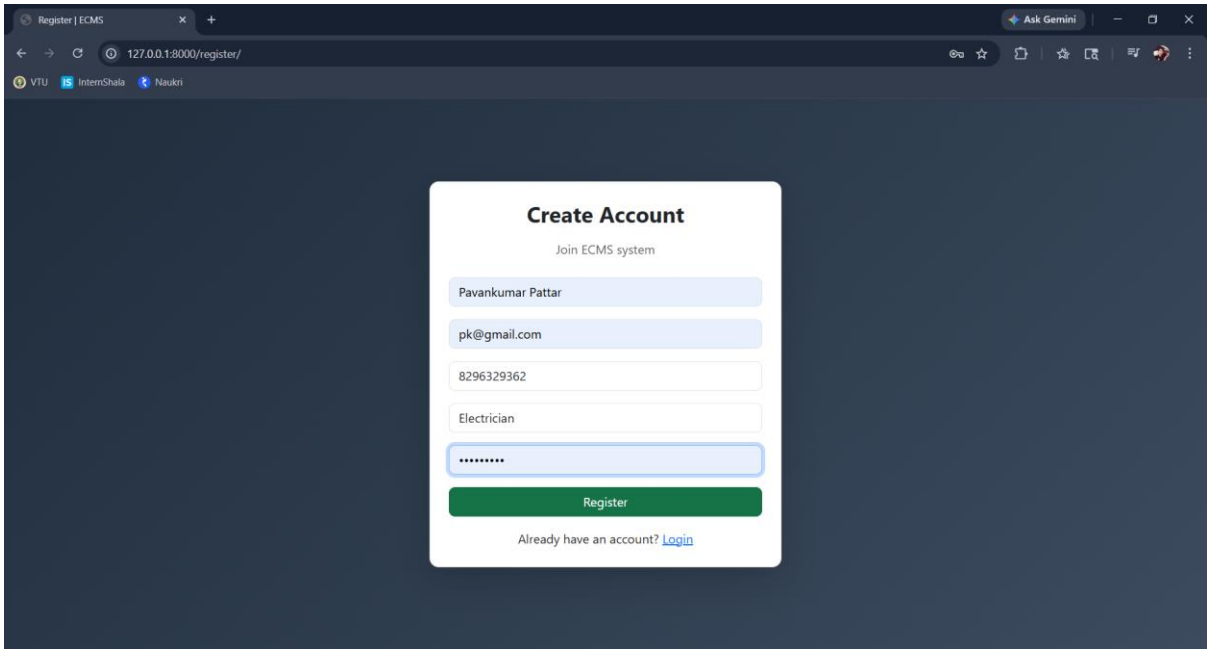
The system includes the following UI modules:

- Home Page

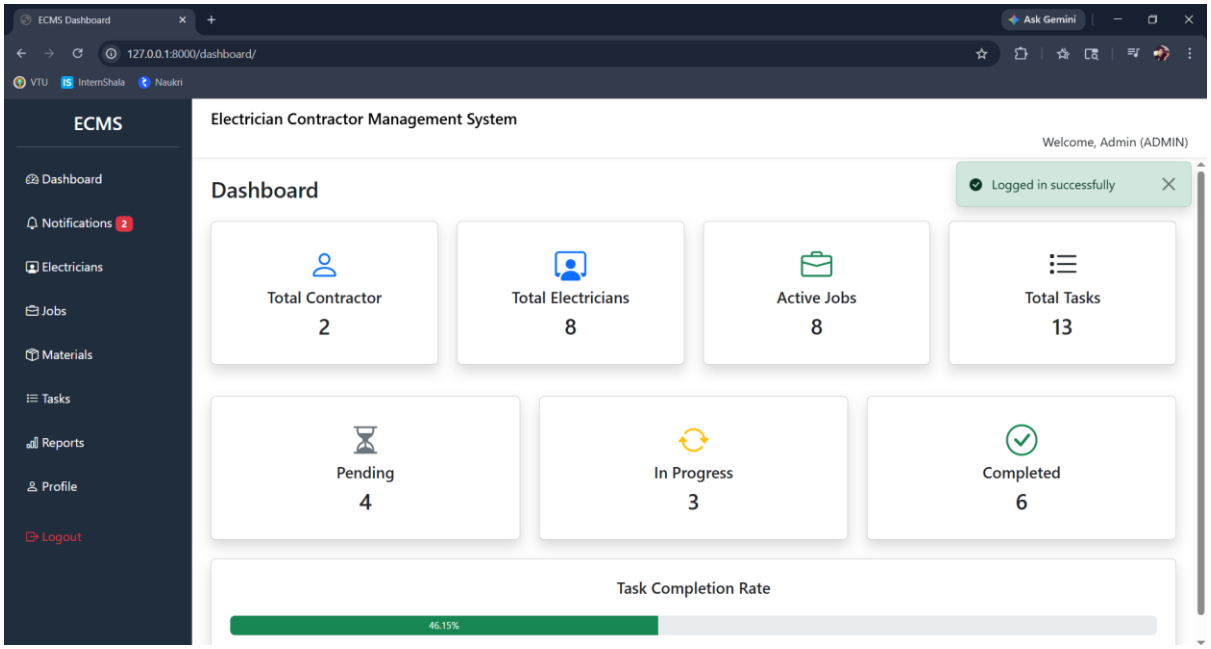


- Login & Registration

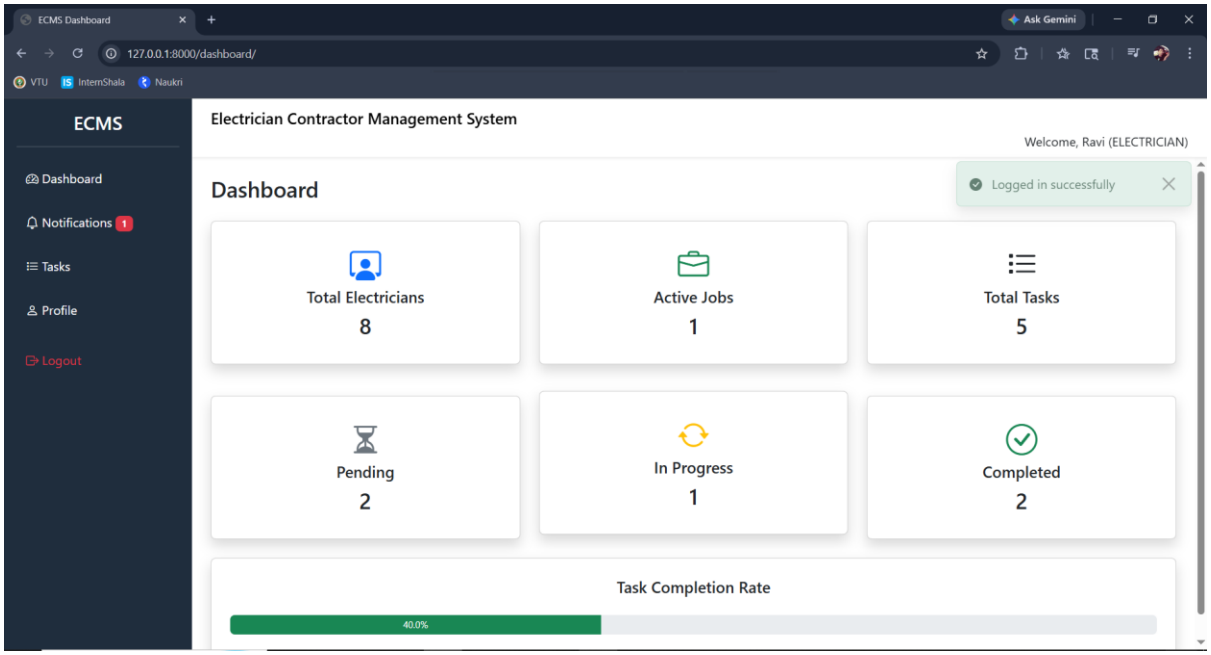




- Admin Dashboard



- Electrician Dashboard



(Screenshots are included in the project README)

4. Project Overview

The Electrician Contractor Management System is a web-based application designed to streamline the management of electricians, contractors, jobs, and tasks within an organization. It provides a centralized platform where administrators can assign jobs and tasks, monitor progress, and maintain structured workflow management. The system reduces manual effort by digitizing operations and ensuring that all activities are tracked efficiently. By integrating role-based access, it allows different users to interact with the system based on their responsibilities.

The platform also enhances communication and coordination through real-time notifications and status updates. Electricians can easily view assigned tasks, update their progress, and meet deadlines effectively,

while administrators receive updates on task completion and overall performance. The dashboard provides useful insights such as task statistics and completion rates, helping in better decision-making. Overall, the system improves productivity, transparency, and accountability within the organization by providing an organized and automated solution.

5. Modules Demonstrated

1. Authentication Module
2. Dashboard Module
3. Electrician Management
4. Job Management
5. Task Management
6. Notification System
7. Reports Module
8. Profile Management

6. Working Demonstration

- User logs in using credentials
- Admin assigns jobs and tasks
- Electrician receives notifications
- Electrician updates task status
- Admin gets notified when tasks are completed
- Dashboard updates dynamically

7. GitHub Link:

https://github.com/Pavankumar1299/Electrician-Contractor-Management-System.git